

SDR based single channel S-AIS Receiver for Satellites using System Generator

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Abstract— The Terrestrial AIS networks have been in operation over the last two decades and helped with the smooth sailing of numerous ships. During this time, the need for information of ships, over a larger footprint seemed necessary, and hence Satellite based AIS systems have been developed. The Satellite-Automatic Identification System (S-AIS) is a critical maritime monitoring system that ensures safety and security of ships by sharing vital information regarding other ships as well as watercrafts in the vicinity. This paper covers the Software Defined Radio (SDR) based receiver designs at the satellite receivers, in order to reliably receive the transmitted AIS message from the ships. A comparison is also drawn between the two receiver algorithms by measuring their Bit Error Rate (BER) performances. The coherent detection based demodulation algorithm proved to be a viable option in terms of both performance and simplicity for the SDR based receiver system. Also this receiver system was tested on a space grade FPGA and the performance was found to be satisfactory.

Keywords—*Satellite Automatic Identification Systems (S-AIS); Self-Organizing Time Division Multiple Access (SOTDMA); Software defined Radio(SDR); Matched filtering demodulation; Delay line correlation demodulation; Additive White Gaussian Noise (AWGN).*

I. INTRODUCTION

Since time immemorial, maritime vessels have always had the need to track each other and make quick decisions based on their positions and other information in order to avoid untoward circumstances. The AIS was developed around the year 2000 to resolve these issues. This system was developed as a traffic collision avoidance system for commercial ships and as a friend or foe identifier in military applications. AIS provides the basic identification information of ship traffic participants within a range of 40 Km. The International Convention for Safety of Life at Sea (SOLAS) describes onboard carriage requirements for the AIS. From 2000, the International Maritime Organization (IMO) has adopted the requirement for all SOLAS ships to be equipped with AIS [1]. According to SOLAS “All ships of 300 gross tonnages and upwards engaged on international voyages and cargo ships of 500 gross tonnages and upwards not engaged on international voyages and passenger ships irrespective of size shall be fitted with an automatic identification system (AIS)” [2].

There are two classes of AIS mobile stations:

- Class A: In order to avoid all the ships of AIS systems speaking at the same time, large ships use an AIS system of class A, which adopts the channel access technique called SOTDMA (Self-Organized TDMA). Class A stations require a maximum transmission power of 12.5W.
- Class B: Smaller vessels fitted with AIS, such as recreational craft, can use a less expensive AIS station of class B, which transmits less frequently. This system uses the CSTDMA (Carrier Sense TDMA) channel access technique. Some old AIS stations of class A can see only the position, but not the identity of these class B stations. Class B stations require a maximum transmission power of 1W.

The AIS standard covers layer 1 to 4 in the Open Systems Interconnection (OSI) Reference Model.

A. Satellite Automatic Identification Systems (S-AIS)

Ship-to-shore and ship-to-ship communication is limited by the horizon. Also, the transmitter antenna altitude located over the ships affects the signal reception over an area surrounding the source ship. Satellite-based AIS is used to overcome the terrestrial VHF coverage limitation with the potential to provide AIS detection service coverage over a larger area on the Earth. When satellites are used to detect AIS signatures then the term Satellite-AIS (S-AIS) is used. In addition, the reception of VHF signals by satellites is proven technology, with many commercial applications.

Satellite reception of these maritime signals can provide a cost-effective tracking, monitoring and messaging capability. With the experience and the proven capability of reception of VHF/UHF signals by satellites, is always a logical step for adapting satellite detection of AIS. The maritime information, gathered from both terrestrial AIS networks as well as Satellite AIS networks, can be merged and analyzed further to provide an increasingly comprehensive overview of ship movements in a given area of interest.

Multiple receiver algorithms have been proposed for the development of an S-AIS receiver to satisfy the conditions of simplicity in design as well as to provide a satisfactory performance. The proposed architectures include the frequency discriminator based demodulator design [3], which provides a good analogue signal processing capability but is

not well suited for digital signal processing in the SDR based environment. Further simplification of the design led to the development of non-coherent demodulation techniques such as the matched filtering demodulation, which is simple and suited for an SDR based environment. But the algorithms failed to provide a good bit error performance and hence called for better coherent demodulation algorithms. One such simple coherent demodulation algorithm is the delay line correlation algorithm. This paper provides simulation results to test and compare the performance of the non-coherent matched filtering algorithm and the coherent delay line correlation demodulation algorithms and also tests the performance of the algorithm on a radiation hardened space grade FPGA using System Generator.

Section II describes the overview of general terrestrial and non-terrestrial AIS systems required to construct a basic AIS transceiver using SDRs. The technical specifications of an AIS signal simulator is discussed in Section III. Two suitable receiver algorithms for this system are discussed in Section IV. The Bit Error performance results from the implementation of these algorithms is shown and analysed in Section V. Section VI deals with the hardware implementation of the algorithm on the space grade Virtex-5 FPGA and the results obtained from the same.

II. AIS FUNCTIONAL OVERVIEW

The AIS message transmission occurs through two VHF channels, one channel at 161.975 MHz frequency and the second channel at 162.025 MHz frequency as specified by the ITU-R AIS Specifications with a transmitted power of 12.5 W [4]. The two channel transmission is done to send redundant information and even if data is lost or corrupted in one channel, the other channel would be available to make sure the data is received. The bandwidth allocated for each channel is 25 kHz. When the data is transmitted, Doppler Shift and other losses are also accounted. Doppler shift lies between -4kHz and 4kHz while the other losses may account for about 3dB loss.

AIS messages are binary messages that can be addressed to either a particular mobile or shore station or they can be broadcasted to all the stations. The AIS Technical Standard ITU-R M.1371-5 lists out 26 different types of AIS messages [4]. The most commonly transmitted messages are of types 1, 2 and 3. The general structure of these types of messages is same as shown in Fig 1. They contain 168 bits of ship information and one frame has 256 bits.

Ramp up	Training sequence	Start flag	Data	FCS	End flag	Buffer
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Fig. 1. Frame Structure of AIS Packet

TABLE I. S-AIS FRAME DESCRIPTION

Packet Data	Length	Functionality
Ramp up	8 bits	It has duration of 1 ms and during this time, the AIS Transmitter switches on.

Training Sequence	24 bits	It is a sequence of alternating ones and zeros and is used for the purpose of synchronization.
Start Flag	8 bits	This is used to indicate that the data would be transmitted after the end of the start flag. It is denoted by "01111110".
Data	168 bits	The actual information that has to be transmitted.
FCS	16 bits	It consists of a CRC-16-CCITT and is used over all the data bits.
End Flag	8 bits	This is used to indicate that all of the data has been transmitted. It is denoted by "01111110", similar to the start flag.
Buffer	24 bits	This is used for the purpose of Bit stuffing, distance delays, repeater delay and jitter.

Apart from the 26 message types, for S-AIS, a new type of message, message 27 was introduced. This contains only 96 bits of ship information and is called the long-range AIS message. When multiple ships are transmitting to the same satellite, they use the Self-Organising Time Division Multiple Access (SOTDMA) method to access the channel. In this variant of TDMA, the ships themselves organize the data transmission. A time slot is pre-booked by a ship for transfer data thus preventing any collisions or any forms of data corruption.

Use of SDRs for the S-AIS application allows the receiver to onboard the satellite to be flexible and allow the adaptation of new receiver technologies on the system without fiddling with the hardware. While the S-AIS payload on the satellite still requires a front end to perform signal reception, and down conversion of the signal to baseband before it can be processed to recover the bits. Thereafter the whole signal processing chain is fed into a suitable FPGA, which performs all the remaining operations for bit recovery.

The transmitted S-AIS signal abides by the frame structure and other requirements of the system. The required transmission power and the available bandwidth serve as major constraints on satellite communication. Often a trade-off is made between the power and bandwidth based on the application. Gaussian Minimum Shift Keying (GMSK) provides the best performance in terms of the required bandwidth and the desired power for the transmission [5]. Doppler shift in the transmitted signal frequency is seen to play a major role and cause discrepancies in the signal reception, while atmospheric attenuation is seen to have minimum-most influence in the VHF band. Therefore, signal losses due to atmospheric attenuation can be neglected at the said frequency range. The system doesn't suffer from multipath fading because of the absence of impediments in its path.

MATLAB v2015b is used to generate the simulated signal from the AIS data bit stream generated by the transmitter onboard the ship. The HDLC-like frame is generated by

adding the other prescribed bits such as the 24-bit training sequence, the start and end flags and the 16-bit CRC check sequence, making the sequence a 256-bit sequence. Further, GMSK modulation is performed on this frame. The output stream after GMSK modulation is sampled at 0.96 Msps. GMSK is performed by Gaussian filtering followed by Minimum Shift Keying (MSK). The signal is transmitted over a 300 kHz (f_c) channel with a bandwidth of 25 kHz and a Doppler shift (f_d) varying from -4 kHz to +4 kHz. If a Doppler shift in the center frequency is fixed at +4 kHz, then the transmitted GMSK signal contains two frequency components corresponding to the two bits transmitted, including 306.4 kHz ($f_c + f_d + 2.4$ kHz) and 301.6 kHz ($f_c + f_d - 2.4$ kHz). Thus, the simplest GMSK demodulator design at the receiver would be a matched filter demodulator at frequencies 306.4 kHz and 301.6 kHz.

III. DESIGN OF RECEIVER ALGORITHMS

A. Matched filtering demodulation

Matched filtering demodulation technique has been traditionally used to receive bits transmitted over a noisy communication channel. Matched filtering is typically used to increase the signal-to-noise ratio at the receiver. This filter is obtained by correlating a known signal, or template, with an unknown signal to detect the presence of the template in the unknown signal. Similarly, the S-AIS receiver involves the multiplication of the incoming signal with two signals of frequencies f_1 (306.4 kHz) and f_2 (301.6 kHz) to detect the presence of these frequencies in the incoming signal. The frequencies f_1 and f_2 correspond to transmission of bits 1 and 0 respectively [6].

Further, the I and the Q channels are obtained by the multiplication of incoming signal with the cosine and sine of the corresponding frequency. The products are then accumulated over an entire bit period. The norm for the products in both the channels are then calculated and the norms are then compared as shown in figure 1. If $z > 0$, 1 is transmitted as the data else 0 is transmitted. Thus. The data bits are recovered with this simple detection technique.

The four generated waveforms for the matched filtering process over one-bit period T_b with a sampling frequency f_s are:

$$\text{Waveform1} = \cos(2\pi f_1 t) \quad (1)$$

$$\text{Waveform2} = -\sin(2\pi f_1 t) \quad (2)$$

$$\text{Waveform3} = \cos(2\pi f_2 t) \quad (3)$$

$$\text{Waveform4} = -\sin(2\pi f_2 t) \quad (4)$$

If $Y(t)$ is the received signal, Y_1 denotes the cross correlation between waveform1 and $Y(t)$.

The two frequencies in the received signal include:

$$f_1 = f_c + f_d + 2.4 \text{ kHz}, f_2 = f_c + f_d - 2.4 \text{ kHz}.$$

$$Y_1(m) = \sum_{n=0}^{N-m-1} (\text{Waveform1})_{n+m} Y_n \quad (5)$$

Similarly, cross correlation is performed with all the generated waveforms to generate Y_1, Y_2, Y_3 and Y_4 as shown in Fig.2.

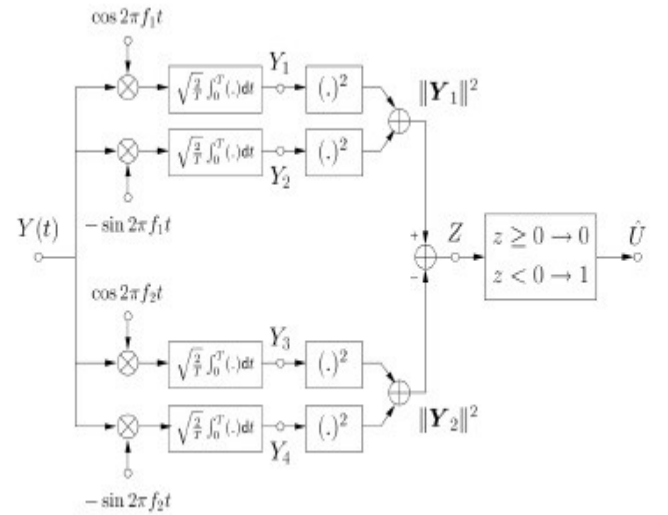


Fig. 2. Matched filter demodulation

Although the non-coherent demodulation technique is simple, widely used and reliable, the bit error performance of the system is not very satisfactory for the application. Hence, a simple coherent demodulation technique has been adopted for the application that gives reliable results and is well suited for an SDR system.

A. Delay line correlation demodulation

The delay line correlator extracts the instantaneous phase and frequency of the incoming signal.

The received signal is sampled at 0.96 Mega Samples per second (Msps) or 100 samples per bit. The signal is first pre-filtered to allow only the S-AIS signal components. The first part of AIS decoding is packet detection, for which a rough FFT is evaluated over thousands of samples or the entire frame length to detect the presence of a packet in the channel. The accurate frequency of the transmitted signal is then obtained by determining the fine FFT of the frame over the training sequence bits only. This is done because the training sequence has an equal number of 1's and 0's and hence the frequency shifts due to these bits are nullified over this range and the detected frequency will be precisely $f_c + f_d$. This detected frequency is then used to down convert the signal and obtain the I and Q channels as shown in Fig. 3. Later low pass filtering is used to remove the higher frequency component after multiplication and is accomplished by accumulation of the samples so as to obtain only 5 samples per bit.

This sample stream is now passed through the delay line correlator. The delay line correlator multiplies a sample with the complex conjugate of a five times delayed sample, in other words it multiplies a sample with the complex conjugate of the corresponding sample in the adjacent bit. The phase of the resulting product is extracted and the instantaneous frequency is obtained. The bit is 1 if the extracted phase is greater than or

equal to 0 and 0 otherwise [6]. The logical bit stream is thus obtained. This bit stream is now oversampled by 5. The final bit can be obtained by extracting the middle sample of every 5 samples representing a bit. The system can be further optimized by obtaining the sample value that occurs maximum number of times in the five samples that represent a bit [7]. The coherent delay line correlator demodulation technique shows considerably better performance than that of the non-coherent matched filtering demodulation.

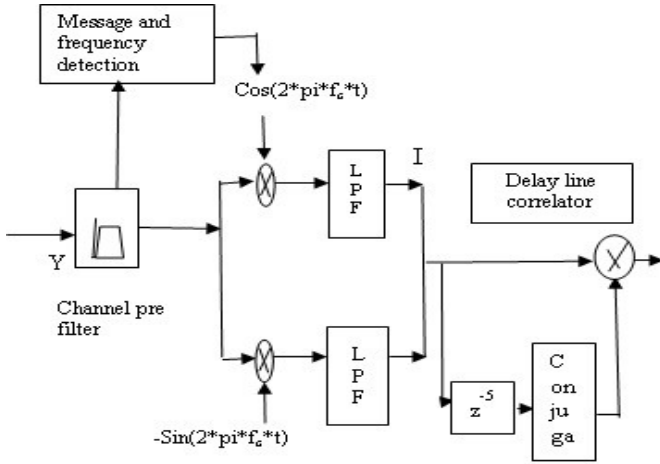


Fig. 3. Delay line correlation demodulation

At the transmitter, the signal is sampled 100 times, which means there are 100 samples per bit (spb). During GMSK modulation, the following operation is performed:

$$\text{Phase (current_sample)} = \text{Phase (previous_sample)} + \text{current_sample_value} * \pi / 2 * \text{spb} \quad (6)$$

If the received sample in the kth time slot is r_k , and has a phase of Φ_k , then the sample will be

$$r_k = \sqrt{E} e^{j\Phi_k} + n_k \quad (7)$$

The product of a sample in the kth and (k-1)th bit will be

$$r_k r_{k-1}^* = E_s e^{j(\Phi_k - \Phi_{k-1})} + \sqrt{E_s} e^{j\Phi_k} n_{k-1}^* + \sqrt{E_s} e^{-j\Phi_{k-1}} n_k + n_k n_{k-1}^* \quad (8)$$

The phase of this product term will be $\Phi_k - \Phi_{k-1}$ which is the phase difference between two samples of corresponding bits, which is $(\text{Phase}(\text{current_sample}) - \text{Phase}(\text{previous_sample}))$.

Thus the current_sample_value(ii) will be

$$\text{Atan}(r_k r_{k-1}^*) * 2 / \pi.$$

IV. RESULTS AND DISCUSSION

MATLAB v2015b is used to obtain the 256 bits S-AIS frame from 168 bits of transmitted data. Further, Gaussian filtering and MSK is performed on the bits to generate the signal at f_c (300kHz), with an added Doppler. Additive White Gaussian Noise (AWGN) is added to the transmitted signal. The Fast Fourier Transform (FFT) of the transmitted signal reveals two signal components f_1 and f_2 at $f_c + f_d + 2.4$ kHz and $f_c + f_d - 2.4$ kHz.

The FFT in Fig. 4 indicates the frequency of the signal as 306.4 kHz (300 kHz + 4 kHz + 2.4 kHz) when a forced Doppler of 4 kHz is added to the signal. The GMSK signal over which Additive white Gaussian noise is added is shown in Fig. 5.

Matched filtering demodulation is performed at the frequencies f_1 and f_2 stated above. The BER versus SNR of the received signal after matched filtering demodulation is as shown in Fig. 6. The figure reveals a decrease in the BER with an increase of E_b/N_0 as expected. The performance of the system is observed and then compared with the delay line correlator demodulation system.

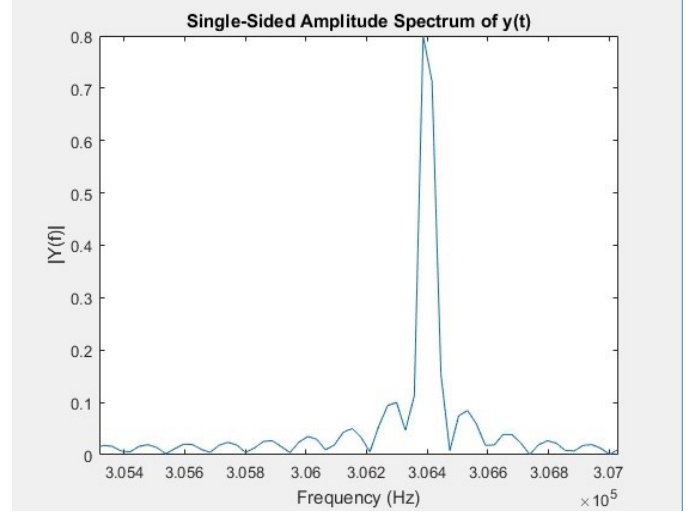


Fig. 4. FFT of transmitted signal

In an alternative demodulation technique, the received signal is bandpass filtered over a bandwidth of $2 * (f_c + 12.5$ kHz + 4 kHz) and the centre frequency of the received signal ($f_c + f_d$) is obtained using FFT. The signal is then downconverted to baseband and the operation of accumulation provides 5 samples per bit. Delay line correlation is performed on these samples and the phase of every complex output sample is obtained. The third sample method or the maximum sample method as discussed is used to recover the decoded bit after which the bits are converted to logical forms. As can be deduced from Fig. 7, for simulation performed on a 256-bit frame over signal to noise ratios of 15 dB to 25 dB the bit error performance of the maximum sample method provides lower bit error rates and hence a better performance.

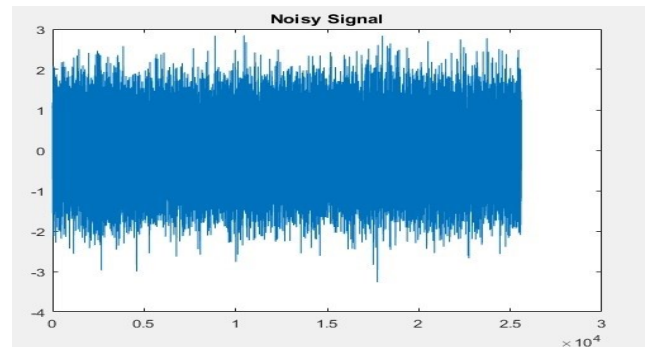


Fig. 5. Noisy GMSK signal

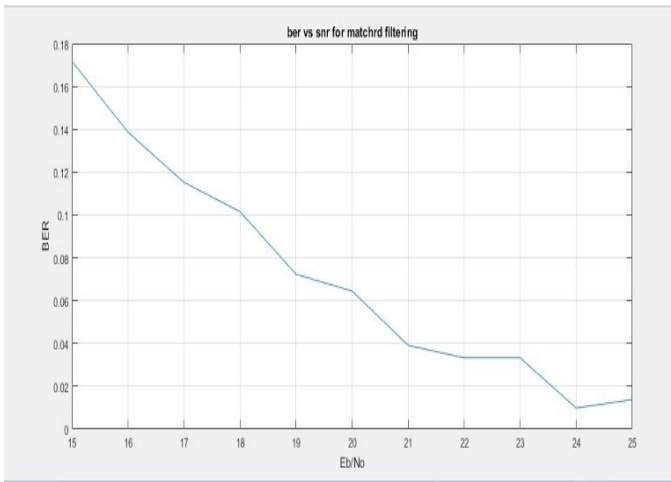


Fig. 6. BER vs SNR of matched filtering demodulation

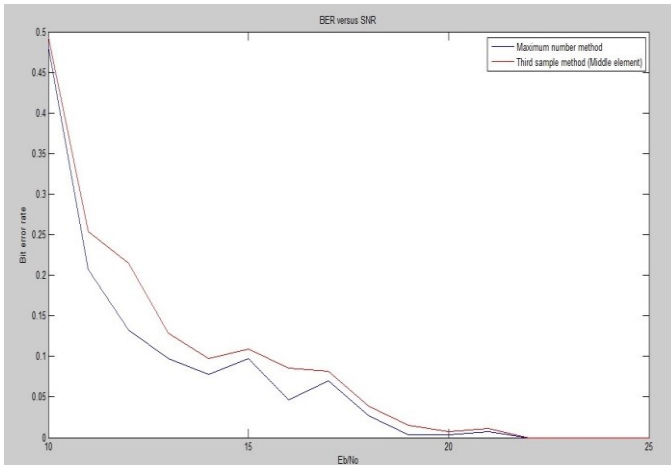


Fig. 7. BER vs SNR of delay line correlation demodulation

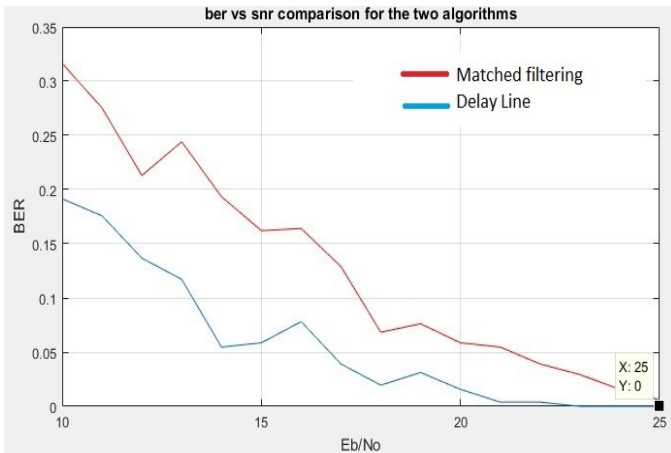


Fig. 8. Performance comparison between the two demodulators

Fig.8. gives a BER comparison of the two algorithms over signal to noise ratios of 10 dB to 25 dB . The blue graph indicates the bit error performance of the delay line correlator while the red line indicates the performance of the matched filter. The delay line correlator evidently performs

significantly better than the matched filter demodulator and is hence a suitable design for the SDR based S-AIS receiver. An approximate gain of 3dB over matched filtering demodulation is shown by the delay line correlator demodulation technique.

V. HARDWARE IMPLEMENTATION

The Virtex-5QV space grade FPGA was used to test the receiver demodulation/decoding algorithm. This FPGA is radiation-hardened by a suitable design technology. Its performance, unparalleled density and tolerance to radiation makes it suitable for space based applications. Easy reconfigurability makes this FPGA a perfect choice for SDR based space applications.

The development of the design is performed on the System Generator (SysGen) foundation by Xilinx. SysGen is a DSP based design tool that works in tandem with Mathworks based Simulink design tool to configure an FPGA. The Simulink tool makes use of several block sets like the modulator, FFT blocks etc. All the other FPGA implementation steps that follow such as compilation, place and route etc are performed automatically to generate an FPGA programming file [8].

In order to verify the receiver system, the system is modelled on the Simulink design environment as shown in Fig.9. The input bit stream is imported from the MATLAB workspace which is the final 256 bits HDLC frame. This data frame is GMSK modulated over Simulink and the upconverted I and Q channels of this data is sent to the receiver. The receiver subsystem is modelled over the SysGen tool as shown in Fig. 10. At the receiver the upconverted I and Q signals are downconverted by multiplication of the incoming waveforms with corresponding cosine and sine signals generated using Direct Digital Synthesis (DDS). The multiplied signal has to be low pass filtered in order to remove the higher frequency component, which is achieved through the process of accumulation. Accumulation is carried out over a period of $20 \cdot T_s$ which in effect is the accumulation of every 20 samples resulting in 5 samples per bit. These samples are further processed over the delay line correlator.

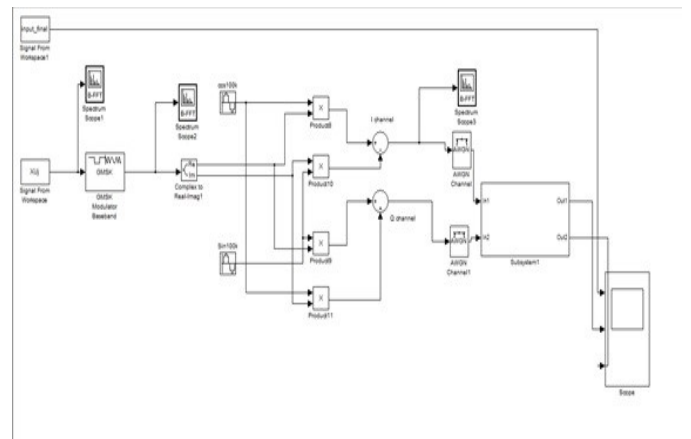


Fig. 9. System model

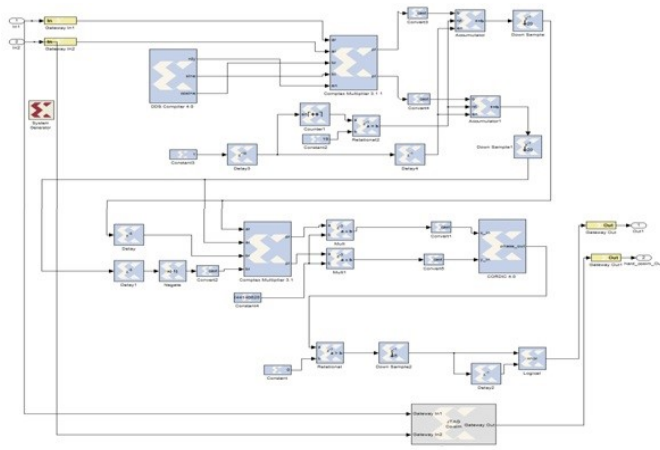


Fig. 10. System generated demodulator subsystem

The delay line correlator multiplies every incoming sample in the sample stream with the complex conjugate of the 5 times delayed sample. In principle, this is the product of sample in a bit with the complex conjugate of the corresponding sample in the adjacent bit. The phase of this product is the phase difference between the two adjacent bits and is extracted using IP Core CORDIC 4.0. The extracted phase/frequency is converted to logical 1 or 0 in order to obtain the logical sampled bit stream. This sample stream is downsampled by 5 and NRZI decoded to give the final 256-bit bitstream. The obtained bitstream is compared with the transmitted bitstream obtained from MATLAB in order to validate the algorithm.

Fig.11 shows the input data stream, the simulator output from hardware co-simulator and the output obtained from the Virtex-5 FPGA at signal to noise ratio of 25 dB. It is seen that the data stream obtained is same as the input bit stream after the explicitly given delay. The results obtained from the space grade FPGA are satisfactory (with 4 error bits in the received signal) and hence confirm the ability of the designed algorithm to work over an SDR based environment.

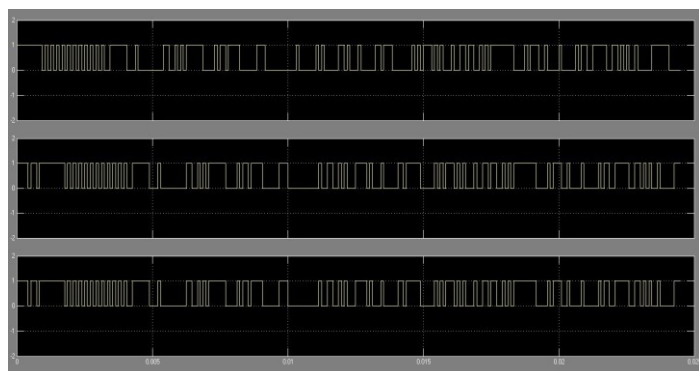


Fig. 11. Input signal, cosimulator output and FPGA output.

VI. CONCLUSION AND FUTURE WORK

S-AIS is undoubtedly the backbone of Safety at Sea and is the only warning system that can forewarn all watercrafts to prevent untoward incidents at sea. Thus the two most important elements in this system would include the presence of a good transmitter that transmits accurate and maximum information about the ship to other ships over large distances and a reliable receiver that obtains accurate information and shares them with ships covering substantial areas. A reliable receiver delay line correlator based receiver algorithm can also be easily supported by the SDR based receiver. This receiver algorithm also shows considerably better performance as compared to the traditional matched filtering based algorithm.

Extensive research in the field of Satellite AIS systems focus on the development of the system and standards for all kind of watercrafts on the sea. This will help in every watercraft to develop a clearer view of all other watercrafts around it and thereby ensuring maximum safety and security. In addition, newer algorithms are required to be developed at the satellite receiver, which allows the decoded bit-stream to be analysed in detail to make suitable deductions and predictions on the behaviour of watercrafts on sea. For instance, the course and speed of two ships on a sea can be used to predict, if they are heading towards a probable cross in their paths using the algorithm.

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